

Image Feature Processing in Deep Learning using Convolutional Neural Networks: An Overview

David H. Hubel and Torsten Wiesel laid the foundations for building the CNN (convolutional neural network) model after their studies and experiments on the human visual cortex. Since then, CNN models have been built with near human accuracy. This article explores image processing with reference to the handling of image features in CNN. It covers the building blocks of the convolution layer, the kernel, feature maps and how the activations are calculated in the convolution layer. It also provides insights into various types of activation feature maps and how these can be used to debug the CNN model to reduce computations and the size of the model.

